2020～2021 学年第 2 学期期末考试试卷

《程序设计原理》（备 A 卷 共 7 页）

（考试时间：2021 年 06 月 18 日）

题号	I	II	III	IV	V	成绩	核分人签字
得分							

Note: All the questions in this exam paper were designed and tested using Code::Blocks with GNU g++ compiler on 32-bit Windows.

I Multiple Choices (20 questions, 20 points)

Instructions:

1. There are 20 multiple choices (1 point each). They are divided into 5 sections. Every section contain 4 questions, and will focus on a single topic.
2. Every question has four different choices. Only one of them is correct. **Choose the right one and write the letter before that choice in the table below. DO NOT write your answers elsewhere.**

1	2	3	4	5	6	7	8	9	10
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**

Question (1) ~ (4) are based on the knowledge of C-style strings and characters.

D (1) What is the storage size of the string literal: `"C:\\CPP2018\\FinalExam.DOC"`

A) 24 B) 25 C) 26 D) 27

C ~~D~~ (2) Which of the following string literals has zero string length, but non-zero storage size?

A) 0 B) “0” C) “” D) ‘\0’

C (3) After evaluating the expression: `*("Hello" + 1)`, what is the datatype and the value of the result?

A) `int, 101` B) `int *, 101` C) `char *, 'e'` D) `char, 'e'`

A (4) `const char *s1 = "ABCD"; const char s2[] = "ABCD";` Which of the following statements will **NOT** result in any error or potential issue?

A) `s1 = "EFGH";` B) `s2 = "EFGH";` C) `s1[0] = 'E';` D) `s2[5] = 'E';`

Question (5) ~ (8) are based on concepts of functions and basic programming skills

(5) Which of the following functions CANNOT be overloaded? B

A) Constructor B) Destructor C) Member function D) Virtual function

(6) Which of the following statements will execute the repetition for at least once? C

A) for B) while C) do ... while D) None of the above

(7) After executing: `int x = 5; int y = x++;` What is the value of x and y respectively?

A) 5, 5 B) 5, 6 C) 6, 6 D) 6, 5 C

(8) Which function will trigger the copy constructor of class Book, when it is called?

A) `void fun(int x, Book *y);` B) `Book &fun(Book *x, int y);` D

C) `void fun(Book &x, int y);` D) `Book *fun(Book x, int *y);`

Question (9) ~ (12) are based on object-oriented programming concepts

(9) Assuming “Book” is a class, the following statement declares three objects: B

```
Book a, b, c;
```

If these objects are local variables within a function, what is the order of destruction if all three objects automatically destruct when the function returned? B

A) a, b, c B) c, b, a C) A random order D) None of the above

(10) Assuming “Book” is a class that has been defined with a null constructor, how many times were the constructor called after executing a statement as follows:

```
Book cpp, math[3], *english[2];
```

D

A) 1 B) 2 C) 3 D) 4

(11) When 初始化 initializing an object, which constructor will be called first? A

A) Base class constructor B) Constructor of the class itself

C) Sub-object class constructor D) Copy constructor of the class itself

(12) Defining a class with the `struct` keyword, what is the default access modifier of its members? B

A) friend B) public C) protected D) private

Question (13)~(16) are based on concepts of data structures and algorithms

(13) When the key sequence {19, 14, 23, 1, 68, 20, 84, 27, 55, 11, 10, 79} are stored using a hash table, the separate chaining method was used to solve collisions. If the hash function is $H(k) = k \% 13$, which keys are stored on the same chain?

A) 23, 10 B) 27, 11 C) 68, 84 D) 79, 55

(14) Which of the following is the closest concept to a LIFO list?

A) Linked-list B) Array C) Queue D) Stack

(15) A binary search tree was constructed using key sequence {50, 30, 10, 15, 35, 55, 53, 33, 37, 62}. After removing node 50, which of the following can be used as the new root node?

A) 37 B) 10 C) 62 D) 15

(16) What is the average time complexity of the insertion sorting algorithm?

A) $O(1)$ B) $O(n)$ C) $O(n^2)$ D) $O(n\log n)$

4) **Object-oriented design.** In object-oriented design, there is a designing pattern, which is called "Singleton". In this designing pattern, a specially designed class cannot be instantiated (实例化) normally. However, by using its static members, this class can be guaranteed to have only one instance during the whole life time of the program. Please read the following program and write the screen output. (4 points)

```
#include <iostream>
using namespace std;
class Singleton {
    public:
        static Singleton &GetObject() {
            if (!m_self)
                m_self = new Singleton();
            else
                m_self->m_CountRefs ++;
            return *m_self;
        }
        bool Dispose() {
            m_CountRefs --;
            if (!m_CountRefs) {
                delete m_self;
                return true;
            }
            return false;
        }
        bool SetValue(int n) { return (n > 0) ? m_ProtectedValue = n : false; }
        int GetValue() { return m_ProtectedValue; }
        int GetCount() { return m_CountRefs; }
    private:
        Singleton() { m_self = this; m_CountRefs = 1; m_ProtectedValue = 0; }
        Singleton(const Singleton &n){}
        Singleton(Singleton &n){}
        ~Singleton() { m_self = 0; m_CountRefs = 0; m_ProtectedValue = 0; }
        int m_CountRefs;
        int m_ProtectedValue;
        static Singleton *m_self;
};
Singleton* Singleton::m_self = 0;
int main() {
    Singleton &a = Singleton::GetObject();
    Singleton &b = Singleton::GetObject();
    Singleton &c = Singleton::GetObject();
    a.SetValue(1);
    b.SetValue(2);
    c.SetValue(a.GetCount());
    cout << a.GetValue() << endl;
    cout << b.GetValue() << endl;
    cout << c.GetValue() << endl;
    cout << c.GetCount() << endl;
    a.Dispose();
    b.Dispose();
    c.Dispose();
    return 0;
}
```

(5) **Thinking in binary.** Please read the following program and write its output. (4 points)

```
#include <iostream>
using namespace std;
unsigned int bitones(unsigned int n) {
    int r = 0;
    while (n) {
        r += n % 2;
        n /= 2;
    }
    return r;
}
int main() {
    cout << bitones(0x00000001) << endl;
    cout << bitones(0x01000100) << endl;
    cout << bitones(0x00000E00) << endl;
    cout << bitones(0x300C0000) << endl;
    return 0;
}
```

Letter Classification. Please read the following program and write its output. (4 points)

```
#include <iostream>
#include <cctype>
using namespace std;
void CountLetters(const char *s)
{
    int c[5] = {};
    for (const char *p = s; *p ; p ++)
        c[
            1 * (isdigit(*p) ? 1 : 0) +
            2 * (islower(*p) ? 1 : 0) +
            3 * (isupper(*p) ? 1 : 0) +
            4 * (ispunct(*p) ? 1 : 0)
        ] ++;
    for (int i = 1 ; i < 5 ; i ++)
        cout << c[i] << endl;
}
int main()
{
    const char *n = "1a2b,c3A4B";
    CountLetters(n);
    return 0;
}
```

Answering table for **II Program reading comprehension**					
(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**

III Briefly answering questions (3 questions, 12 points)

Instructions: Write your answers briefly in the blanks below every question.

(1) **Huffman Tree.** Given a Huffman encoding scheme(方案) {a: 0, b:111, c:1011, d: 100, e: 110, f:1010}, answer the following questions:

i. Draw the Huffman Tree below. Internal nodes can be left blank. (2 points)

ii. Which letter has the highest appearance frequency? (2 points)

(2) **Binary Tree.** Given a binary tree with n_0 leaf nodes and k edges. Answer the following questions.

i. How many nodes are there in the binary tree? (2 points)

ii. How many nodes have 2 children? (2 points)

Binary Tree Traversal (遍历). The post-order traversal of a binary tree is DHIEBFGCA, while its in-order traversal DBHEIAFCG.

i. Please draw the tree structure below. (3 points)

ii. What is the pre-order traversal of this binary tree? (1 points)

IV Programming Cloze Test (16 blanks, 32 points)

Instructions: Read the following programs very carefully and fill in the blanks. Write your answers in the answering table on **PAGE A6. DO NOT write your answers elsewhere.**

(1) **Selection sort.** Selection sort is a sorting algorithm that can be applied on an array. The following function char *strsort (char *) will sort all the letters in a C-style string in an **ascending order (升序)**; and return the sorted string. The parameter s is a C-style string that is to be sorted. In this question, we assume that the C-style string s is writable. We also assume that only lower case letters appear in s. Please fill in the blanks in the following function. (8 points)

```
char *strsort(char *s)
{
    for (char *p = ___(1)___; *p; p ++)
        for (char *q = p + 1; *q; ___(2)___)
            if (___(3)___)
            {
                char t = *p; *p = *q;___(4)___;
            }
    return s;
}
```

(2) **File manipulation.** The following program opens a file with name "D:\Data.txt". It writes 100 integers, from 0 to 99, into the file. At last, it writes the summation (和) of all 100 integers into the same file, before the file is closed. Every integer will occupy a single line in the file, including the summation. Please fill in the blanks in the following program. (8 points)

```
#include <iostream>
#include <fstream>
using namespace std;
int main()
{
    ___(5)___ ofs("d:\\Data.txt");
    if (!ofs)
        return 0;
    int s = 0;
    for (int i = 0 ; i < 100 ; i ++)
    {
        ofs << ___(6)___ << endl;
        ___(7)___ += i;
    }
    ofs << s << endl;
    ofs.___(8)___;
}
```

(3) **Palindrome Number (回文数).** The following function determines whether the integer n is a palindrome number. A palindrome number is a number that is identical to its digital reversed number. For example, the number 13031 is a palindrome number, while the number 12345 is not a palindrome number. Please fill in the blanks in the following function. (8 points)

```
bool IsPalindrome(int n)
{
    int ___(9)___, d = n;
    for (t = ___(10)___; n ; n /= 10)
        t = ___(11)___ + n % 10;
    return t == ___(12)___;
}
```

(4) **Linked list as a data container.** The class String is a linked list, whose data elements are defined as class Char. As shown in the constructor of the class String, the linked list has a header node that is not used for storing data. Function Display print all data elements as a string on the screen, Function DelChar removes a specific character from the list. You should read this class carefully and fill in the blanks. (8 points)

```
struct Char {
    char c;
    Char *next;
};
class String {
public:
    String(const char *s) : head(0), Length(0) {
        head = new Char();
        head->c = 0; head->next = 0;
        Char *n = head;
        for (const char *p = s ; *p ; p ++) {
            Char *t = new Char();
            t->c = *p; t->next = 0;
            n->next = ___(13)___; n = n->next;
            Length ++;
        }
    }
    ~String() {
        Char *p = head;
        while (p->next) {
            Char *q = p->next;
            p->next = q->next;
            delete ___(14)___; q = 0;
        }
    }
    void Display();
    void DelChar(char c);
private:
    Char *head;
    int *Length;
};
```

```
void String::Display() {
    Char *p = head->next;
    while (p) {
        cout << p->c;
        p = ___(15)___;
    }
}
void String::DelChar(char c) {
    Char *p = head;
    while (p->next) {
        Char *q = p->next;
        if (___(16)___ == c) {
            p->next = q->next;
            delete q; q = 0;
        }
        else {
            p = p->next; q = 0;
        }
    }
}
```

Answering table for IV Programming cloze test							
1		2		3		4	
5		6		7		8	
9		10		11		12	
13		14		15		16	

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Note: All the questions in this exam paper were designed and tested using Code::Blocks with GNU g++ compiler on 32-bit Windows.

I. Multiple Choices (20 questions, 20 points)

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Question (1) ~ (4) are based on concepts of functions and basic programming skills

(1) Which of the following functions **CANNOT** be overloaded?

A) Constructor B) Destructor C) Member function D) Virtual function

(2) Which of the following statements will execute the repetition for at least once?

A) for B) while C) do … while D) None of the above

(3) After executing: `int x = 5; int y = x++;` What is the value of x and y respectively?

A) 5, 5 B) 5, 6 C) 6, 6 D) 6, 5

(4) Which function will trigger the copy constructor of class Book, when it is called?

A) `void fun(int x, Book *y);` B) `Book &fun(Book *x, int y);`

C) `void fun(Book &x, int y);` D) `Book *fun(Book x, int *y);`

Question (5) ~ (8) are based on the knowledge of C-style strings and characters.

(5) What is the storage size of the string literal: `"C:\\CPP2018\\FinalExam.DOC"`

A) 24 B) 25 C) 26 D) 27

(6) Which of the following string literals has zero string length, but non-zero storage size?

A) 0 B) "0" C) "" D) "\0"

(7) After evaluating the expression: `*("Hello" + 1)`, what is the datatype and the value of the result?

A) `int, 101` B) `int *, 101` C) `char *, 'e'` D) `char, 'e'`

(8) `const char *s1 = "ABCD"; const char s2[] = "ABCD";` Which of the following statements will **NOT** result in any error or potential issue?

A) `s1 = "EFGH";` B) `s2 = "EFGH";` C) `s1[0] = 'E';` D) `s2[5] = 'E';`

Question (9) ~ (12) are based on the following program:

```
#include <iostream>
using namespace std;
int main() {
    int n[10][10] = {};
    int (*pa)[100] = (int(*)[100])n;
    int (*pb)[20] = (int(*)[20])n;
    for (int i = 0; i < 100; i ++)
        (*pa)[i] = i;
    cout << pb[3][1] << endl;
    return 0;
}
```

(9) What is the output of this program:

A) 59 B) 60 C) 61 D) 62

(10) Which of the following statement best describes the datatype of identifier pa and n

A) pa is an integer; and n is an array.

B) pa is an integer, and n is also an integer.

C) pa is a row pointer; and n is an array.

D) pa is a row pointer; and n is an integer.

(11) What is the value of `sizeof(n)` and `sizeof(pa)`

A) 400, 400 B) 400, 4 C) 4, 400 D) 4, 4

(12) What will happen if we replace `(*pa)[i] = i;` with `*pa[i] = i;`

A) Run-time error. B) Compile-time error.

C) Nothing will happen D) Output will be changed.

Question (13) ~ (16) are based on object-oriented programming concepts

(13) Defining a class with the `struct` keyword, what is the default access modifier of its members?

A) friend B) public C) protected D) private

(14) When initializing an object, which constructor will be called first?

A) Base class constructor B) Constructor of the class itself

C) Sub-object class constructor D) Copy constructor of the class itself

(15) Assuming "Book" is a class, the following statement declares three objects:

```
Book a, b, c;
```

If these objects are local variables within a function, what is the order of destruction if all three objects automatically destruct when the function returned?

A) a, b, c B) c, b, a C) A random order D) None of the above

(16) Assuming "Book" is a class that has been defined with a null constructor, how many times were the constructor called after executing a statement as follows:

```
Book cpp, math[3], *english[2];
```

A) 1 B) 2 C) 3 D) 4

Question (17)~(20) are based on concepts of data structures and algorithms

(17) When the key sequence {19, 14, 23, 1, 68, 20, 84, 27, 55, 11, 10, 79} are stored using a hash table, the separate chaining method was used to solve collisions. If the hash function is $H(k) = k \% 13$, which keys are stored on the same chain?

A) 23, 10 B) 27, 11 C) 68, 84 D) 79, 55

(18) A binary search tree was constructed using key sequence {50, 30, 10, 15, 35, 55, 53, 33, 37, 62}. After removing node 50, which of the following can be used as the new root node?

A) 37 B) 10 C) 62 D) 15

(19) Which of the following is the closest concept to a LIFO list?

A) Linked-list B) Array C) Queue D) Stack

(20) What is the average time complexity of the insertion sorting algorithm?

A) $O(1)$ B) $O(n)$ C) $O(n^2)$ D) $O(n\log n)$

II Program reading comprehension (6 questions, 24 points)

Instructions: Read each of the following program, write the screen output in the answering table on **PAGE B4. DO NOT write your answers elsewhere.**

(1) **Recursion.** Please read the following program and write its output. (4 points)

```
#include <iostream>
using namespace std;
int oeis45(int n) {
    if (n <= 1)
        return n;
    else
        return oeis45(n - 1) + oeis45(n - 2);
}
int main() {
    for (int i = 2 ; i < 6 ; i ++)
        cout << oeis45(i) << endl;
    return 0;
}
```

(2) **Static variable.** Please read the following program and write its output. (4 points)

```
#include <iostream>
using namespace std;
void Increament() {
    static int n = 0;
    n ++; cout << n << endl;
}
int main() {
    for (int i = 0 ; i < 4 ; i ++)
        Increament();
    return 0;
}
```

(3) **Letter Classification.** Please read the following program and write its output. (4 points)

```
#include <iostream>
#include <cctype>
using namespace std;
void CountLetters(const char *s) {
    int c[5] = {};
    for (const char *p = s; *p ; p ++)
        c[
            1 * (isdigit(*p) ? 1 : 0) +
            2 * (islower(*p) ? 1 : 0) +
            3 * (isupper(*p) ? 1 : 0) +
            4 * (ispunct(*p) ? 1 : 0)
        ] ++;
    for (int i = 1 ; i < 5 ; i ++)
        cout << c[i] << endl;
}
int main() {
    const char *n = "1a2b,c3A4B";
    CountLetters(n);
    return 0;
}
```

(4) **Object-oriented design.** In object-oriented design, there is a designing pattern, which is called "Singleton". In this designing pattern, a specially designed class cannot be instantiated (实例化) normally. However, by using its static members, this class can be guaranteed to have only one instance during the whole life time of the program. Please read the following program and write the screen output. (4 points)

```
#include <iostream>
using namespace std;
class Singleton {
    public:
        static Singleton &GetObject() {
            if (!m_self)
                m_self = new Singleton();
            else
                m_self->m_CountRefs ++;
            return *m_self;
        }
        bool Dispose() {
            m_CountRefs --;
            if (!m_CountRefs) {
                delete m_self;
                return true;
            }
            return false;
        }
        bool SetValue(int n) { return (n > 0) ? m_ProtectedValue = n : false; }
        int GetValue() { return m_ProtectedValue; }
        int GetCount() { return m_CountRefs; }
    private:
        Singleton() { m_self = this; m_CountRefs = 1; m_ProtectedValue = 0; }
        Singleton(const Singleton &n){}
        Singleton(Singleton &n){}
        ~Singleton() { m_self = 0; m_CountRefs = 0; m_ProtectedValue = 0; }
        int m_CountRefs;
        int m_ProtectedValue;
        static Singleton *m_self;
};
Singleton* Singleton::m_self = 0;
int main() {
    Singleton &a = Singleton::GetObject();
    Singleton &b = Singleton::GetObject();
    Singleton &c = Singleton::GetObject();
    a.SetValue(1);
    b.SetValue(2);
    c.SetValue(a.GetCount());
    cout << a.GetValue() << endl;
    cout << b.GetValue() << endl;
    cout << c.GetValue() << endl;
    cout << c.GetCount() << endl;
    a.Dispose();
    b.Dispose();
    c.Dispose();
    return 0;
}
```

(5) **Thinking in binary.** Please read the following program and write its output. (4 points)

```
#include <iostream>
using namespace std;
unsigned int bitones(unsigned int n) {
    int r = 0;
    while (n) {
        r += n % 2;
        n /= 2;
    }
    return r;
}
int main() {
    cout << bitones(0x00000001) << endl;
    cout << bitones(0x01000100) << endl;
    cout << bitones(0x00000E00) << endl;
    cout << bitones(0x300C0000) << endl;
    return 0;
}
```

(6) **Word Count.** Please read the following program and write its output. (4 points)

```
#include <iostream>
using namespace std;
int WordCount(const char *s);
int main() {
    const char *Test[] = { "See you next year", "Happy new year",
        "Hello    World", "Failed  \0   again and again." };
    int n = sizeof(Test) / sizeof(Test[0]);
    for (int i = 0 ; i < n ; i ++)
        cout << WordCount(Test[i]) << endl;
    return 0;
}
int WordCount(const char *s) {
    int r = 0;
    const char *p = s;
    while (*p != '\0') {
        r++;
        while (*p != ' ' && *p != '\0') p++;
        while (*p == ' ') p++;
    }
    return r;
}
```

Answering table for II Program reading comprehension

(1)	(2)	(3)	(4)	(5)	(6)

III Briefly answering questions (3 questions, 12 points)

Instructions: Write your answers briefly in the blanks below every question.

(1) **Binary Tree.** Given a binary tree with n_0 leaf nodes and k edges. Answer the following questions.

i. How many nodes are there in the binary tree? (2 points)

ii. How many nodes have 2 children? (2 points)

(2) **Huffman Tree.** Given a Huffman encoding scheme(方案) {a: 0, b:111, c:1011, d: 100, e: 110, f:1010}, answer the following questions:

i. Draw the Huffman Tree below. Internal nodes can be left blank. (2 points)

ii. Which letter has the highest appearance frequency? (2 points)

Binary Tree Traversal (遍历). The post-order traversal of a binary tree is DHIEBFGCA, while its in-order traversal DBHEIAFCG.

i. Please draw the tree structure below. (3 points)

ii. What is the pre-order traversal of this binary tree? (1 points)

IV Programming Cloze Test (16 blanks, 32 points)

Instructions: Read the following programs very carefully and fill in the blanks. Write your answers in the answering table on **PAGE B6. DO NOT write your answers elsewhere.**

(1) Palindrome Number (回文数). The following function determines whether the integer n is a palindrome number. A palindrome number is a number that is identical to its digital reversed number. For example, the number 13031 is a palindrome number, while the number 12345 is not a palindrome number. Please fill in the blanks in the following function. (8 points)

```
bool IsPalindrome(int n)
{
    int ___(1)___, d = n;
    for (t = ___(2)___; n ; n /= 10)
        t = ___(3)___ + n % 10;
    return t == ___(4)___;
}
```

(2) Linked list as a data container. The class String is a linked list, whose data elements are defined as class Char. As shown in the constructor of the class String, the linked list has a header node that is not used for storing data. Function Display print all data elements as a string on the screen, Function DelChar removes a specific character from the list. You should read this class carefully and fill in the blanks. (8 points)

```
struct Char {
    char c;
    Char *next;
};
class String {
public:
    String(const char *s) : head(0), Length(0) {
        head = new Char();
        head->c = 0; head->next = 0;
        Char *n = head;
        for (const char *p = s ; *p ; p ++) {
            Char *t = new Char();
            t->c = *p; t->next = 0;
            n->next = ___(5)___; n = n->next;
            Length ++;
        }
    }
    ~String() {
        Char *p = head;
        while (p->next) {
            Char *q = p->next;
            p->next = q->next;
            delete ___(6)___; q = 0;
        }
    }
    void Display();
    void DelChar(char c);
private:
    Char *head;
    int *Length;
};
```

```
void String::Display() {
    Char *p = head->next;
    while (p) {
        cout << p->c;
        p = ___(7)___;
    }
}
void String::DelChar(char c) {
    Char *p = head;
    while (p->next) {
        Char *q = p->next;
        if (___(8)___ == c) {
            p->next = q->next;
            delete q; q = 0;
        }
        else {
            p = p->next; q = 0;
        }
    }
}
```

(3) **Selection sort.** Selection sort is a sorting algorithm that can be applied on an array. The following function char *strsort (char *) will sort all the letters in a C-style string in an **ascending order (升序)**; and return the sorted string. The parameter s is a C-style string that is to be sorted. In this question, we assume that the C-style string s is writable. We also assume that only lower case letters appear in s. Please fill in the blanks in the following function. (8 points)

```
char *strsort(char *s)
{
    for (char *p = ___(9)___; *p; p ++)
        for (char *q = p + 1; *q; ___(10)___)
            if (___(11)___)
            {
                char t = *p; *p = *q;___(12)___;
            }
    return s;
}
```

(4) **File manipulation.** The following program opens a file with name "D:\Data.txt". It writes 100 integers, from 0 to 99, into the file. At last, it writes the summation (和) of all 100 integers into the same file, before the file is closed. Every integer will occupy a single line in the file, including the summation. Please fill in the blanks in the following program. (8 points)

```
#include <iostream>
#include <fstream>
using namespace std;
int main()
{
    ___(13)___ ofs("d:\\Data.txt");
    if (!ofs)
        return 0;
    int s = 0;
    for (int i = 0 ; i < 100 ; i ++)
    {
        ofs << ___(14)___ << endl;
        ___(15)___ += i;
    }
    ofs << s << endl;
    ofs.___(16)___;
}
```

Answering table for IV Programming cloze test

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V Programming (1 question, 12 points)

In this part, you are required to **write a complete program**. This program includes two functions. **You MUST implement BOTH of them.** One function is the main function, the other function is called "ReverseVowels". The prototype of this "ReverseVowels" can be found below, which is NOT allowed to be altered. This function takes only one parameter, the C-style string s. The returned value of this function is also a C-style string, which is the string s after reversing the order of all vowels (元音) letters. For example: if s is "hello", then the function should return "holle". For another example: if s is "pass this exam", the function should return "pass thes ixam". **Only letter 'a', 'e', 'i', 'o', and 'u' are recognized as vowel letters, both upper and lower cases.** In the function body of "ReverseVowels", you are not allowed either to define arrays or to dynamically allocate memory. In the main function, you should use cin to read an input string from keyboard, use the function ReverseVowels to process this string and write the result on screen using cout. In the function body of main, you are allowed to define arrays or to allocate heap memory blocks. If you allocate heap memory blocks, you must release them before the termination of the program to prevent memory leak. You program should be able to read a string no longer than 1023 characters. Only library functions in the following header files are allowed: <iostream>, <cctype> and <cstring>. Using any library functions that are not allowed will result in zero score of this question. **However, you are allowed to define your own functions other than the required two functions.**

The prototype of ReverseVowels function:

```
char *ReverseVowels(char *s);
```

Please write your entire program below. First few lines are given, which are NOT allowed to be altered.

```
#include <iostream>
#include <cctype>
#include <cstring>
using namespace std;

char *ReverseVowels(char *s);
// Your codes go from here
```